

BIOLOGY MASTER SYLLABUS

CLASS 12 • NCERT / CBSE

Chapter 1: Sexual Reproduction in Flowering Plants

- Flower Structure
- Pre-fertilisation Events
- Microsporogenesis
- Megasporogenesis
- Pollination
- Double Fertilisation
- Post-fertilisation Events
- Endosperm Development
- Embryo Development
- Seed Formation
- Fruit Formation
- Apomixis
- Polyembryony

Chapter 2: Human Reproduction

- Male Reproductive System
- Female Reproductive System
- Gametogenesis
- Menstrual Cycle
- Fertilisation
- Cleavage
- Blastocyst Formation
- Implantation
- Pregnancy
- Embryonic Development
- Parturition
- Lactation

Chapter 3: Reproductive Health

- Reproductive Health
- Population Explosion
- Birth Control
- Contraceptive Methods
- Medical Termination of Pregnancy
- Sexually Transmitted Infections
- Infertility
- Assisted Reproductive Technologies

Chapter 4: Principles of Inheritance and Variation

- Mendelian Genetics
- Monohybrid Cross
- Dihybrid Cross
- Laws of Inheritance
- Incomplete Dominance
- Codominance
- Multiple Alleles
- Blood Groups
- Pleiotropy
- Chromosomal Theory of Inheritance
- Linkage
- Crossing Over
- Sex Determination
- Genetic Disorders

- Pedigree Analysis

Chapter 5: Molecular Basis of Inheritance

- DNA
- RNA
- DNA Structure
- DNA Replication
- Transcription
- Genetic Code
- Translation
- Gene Expression
- Lac Operon
- Human Genome Project
- DNA Fingerprinting

Chapter 6: Evolution

- Origin of Life
- Evolutionary Theories
- Evidences of Evolution
- Adaptive Radiation
- Biological Evolution
- Natural Selection
- Hardy-Weinberg Principle
- Genetic Drift
- Speciation
- Human Evolution

Chapter 7: Human Health and Disease

- Human Health
- Pathogens
- Immunity
- Immune System
- Vaccination
- Allergy
- Autoimmunity
- AIDS
- Cancer
- Drug and Alcohol Abuse

Chapter 8: Microbes in Human Welfare

- Microbes in Household Products
- Microbes in Industrial Products
- Sewage Treatment
- Biogas Production
- Biocontrol Agents
- Biofertilisers

Chapter 9: Biotechnology: Principles and Processes

- Biotechnology
- Genetic Engineering

- Recombinant DNA Technology
- Restriction Enzymes
- DNA Ligase
- Cloning Vectors
- Competent Host
- Polymerase Chain Reaction
- Gene Cloning
- Downstream Processing

Chapter 10: Biotechnology and Its Applications

- Biotechnology in Agriculture
- Genetically Modified Crops
- Pest Resistant Crops
- Biotechnology in Medicine
- Recombinant Therapeutics
- Gene Therapy
- Molecular Diagnosis
- ELISA
- Transgenic Animals
- Biopatents
- Biosafety
- Bioethics